

By Theorem 3 from Chapter 1, the matrix equation $Ly=b$ has the same solution as the system of equations corresponding to the augmented matrix $[L \ b]$.

$$\begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ -1 & 1 & 0 & 0 & 5 \\ 2 & -5 & 1 & 0 & 7 \\ -3 & 8 & 3 & 1 & 11 \end{bmatrix} \xrightarrow[\text{+ Row 1}]{\text{Row 2 = Row 2}} \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 2 & -5 & 1 & 0 & 7 \\ -3 & 8 & 3 & 1 & 11 \end{bmatrix} \xrightarrow[-2 \times \text{Row 1}]{\text{Row 3 = Row 3}} \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & -5 & 1 & 0 & 25 \\ -3 & 8 & 3 & 1 & 11 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & -5 & 1 & 0 & 25 \\ -3 & 8 & 3 & 1 & 11 \end{bmatrix} \xrightarrow[\text{+ } 3 \times \text{Row 1}]{\text{Row 4 = Row 4}} \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & -5 & 1 & 0 & 25 \\ 0 & 8 & 3 & 1 & -16 \end{bmatrix} \xrightarrow[\text{+ } 5 \times \text{Row 2}]{\text{Row 3 = Row 3}} \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 8 & 3 & 1 & -16 \end{bmatrix}$$

$$\xrightarrow[\text{8} \times \text{Row 2}]{\text{Row 4 = Row 4 -}} \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 3 & 1 & 16 \end{bmatrix} \xrightarrow[\text{- } 3 \times \text{Row 3}]{\text{Row 4 = Row 4}} \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\therefore y = \begin{bmatrix} -9 \\ -4 \\ 5 \\ 1 \end{bmatrix} \quad Ux = y \Rightarrow \begin{bmatrix} 3 & -7 & -2 & 2 \\ 0 & -2 & -1 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix} x = \begin{bmatrix} -9 \\ -4 \\ 5 \\ 1 \end{bmatrix}$$

By Theorem 3 from Chapter 1, the matrix equation $Ux=y$ has the same solution as the system of equations corresponding to the augmented matrix $[U \ y]$.

$$\begin{bmatrix} 3 & -7 & -2 & 2 & -9 \\ 0 & -2 & -1 & 2 & -4 \\ 0 & 0 & -1 & 1 & 5 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix} \xrightarrow[\text{+ } 2 \times \text{Row 4}]{\text{Row 1 = Row 1}} \begin{bmatrix} 3 & -7 & -2 & 0 & -7 \\ 0 & -2 & -1 & 2 & -4 \\ 0 & 0 & -1 & 1 & 5 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix}$$

$$\xrightarrow[\text{2} \times \text{Row 4}]{\text{Row 2 = Row 2 +}} \begin{bmatrix} 3 & -7 & -2 & 0 & -7 \\ 0 & -2 & -1 & 0 & -2 \\ 0 & 0 & -1 & 1 & 5 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix} \xrightarrow[\text{+ Row 4}]{\text{Row 3 = Row 3}} \begin{bmatrix} 3 & -7 & -2 & 0 & -7 \\ 0 & -2 & -1 & 0 & -2 \\ 0 & 0 & -1 & 0 & 6 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix}$$

$$\xrightarrow[\times (-1)]{\text{Row 4 = Row 4}} \begin{bmatrix} 3 & -7 & -2 & 0 & -7 \\ 0 & -2 & -1 & 0 & -2 \\ 0 & 0 & -1 & 0 & 6 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \xrightarrow[\text{+ Row 3}]{\text{Row 2 = Row 2}} \begin{bmatrix} 3 & -7 & 0 & 0 & -19 \\ 0 & -2 & 0 & 0 & -8 \\ 0 & 0 & 1 & 0 & 6 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

$$\xrightarrow[\text{+ } 2 \times \text{Row 3}]{\text{Row 1 = Row 1}} \begin{bmatrix} 3 & -7 & 0 & 0 & -19 \\ 0 & -2 & 0 & 0 & -8 \\ 0 & 0 & 1 & 0 & 6 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$